

ZEROPAY DIGITAL SERVICES LTD

United Kingdom Subsidiary of BETONIQ WEST LTD (RC 1496603, Nigeria)

TriageAI — Technical White Paper

SMS/USSD symptom triage, clinic routing, and blood-panel risk screening.

Prepared by: Tom Ugbodaga — Shareholder & CEO

MSc Computer Systems Security (University of Gloucestershire — England, 2011)

LinkedIn: <https://www.linkedin.com/in/tom-deggs-ugbodaga-2306692bb>

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Executive Summary

TriageAI is a Zeropay Digital Services Ltd venture (UK subsidiary of BETONIQ WEST LTD) — SMS/USSD symptom triage, clinic routing, and blood-panel risk screening.

Key statistic: Millions of people die annually from cancers that are highly treatable when caught early, yet roughly half the world's population lacks access to basic diagnostic services (WHO).

Problem Statement

In low-connectivity and low-income regions, patients often have no accessible way to determine whether a symptom warrants urgent care, and existing blood test results are frequently filed without being reviewed for early risk markers due to clinician time pressure. TriageAI addresses both the front-door triage gap and the underused-data gap.

Technical Architecture

- SMS/USSD gateway enables symptom intake with zero smartphone or data requirement.
- Rule-based classifier maps reported symptoms to an urgency tier and routes to the nearest capable clinic.
- Premium blood-panel module compares patient-entered marker values against established clinical reference ranges and flags out-of-range results for specialist follow-up.
- All risk-flag outputs are explicitly labeled as a screening aid, never a diagnosis — by design, to remain outside full medical-device regulatory classification while still delivering real triage value.

Core Data Model

Entity	Purpose
Patient	Anonymized or consented patient profile and contact channel.
SymptomReport	Reported symptoms, urgency classification, timestamp.
ClinicDirectory	Verified clinic locations, capacity, contact details.
BloodPanelUpload	Patient-entered blood marker values (CBC, PSA, CEA, CA-125, AFP, etc.).
RiskFlag	Rule-based/ML-assisted flag against reference ranges, always paired with a specialist-referral disclaimer.

Security, Privacy & Compliance

- Health data encrypted at rest and in transit; access scoped strictly to the reporting patient and their nominated clinic.
- No blood-panel data used for anything beyond generating the immediate risk-flag output — no secondary use without explicit consent.
- Clear regulatory positioning as clinical decision support / screening aid, distinct from a diagnostic medical device, to align with a realistic and lawful path to market.

Feasibility & Build Status

Symptom-intake and clinic-routing logic live on Hugging Face Spaces; blood-panel reference-range comparison logic implemented as a rule-based screening aid.

Market Opportunity

Digital health market in Sub-Saharan Africa projected to reach \$3.5B by 2030. Target beachhead: rural primary health centres lacking specialist access.

Grant & Funder Fit

Strong fit for global health innovation funding: Gates Foundation Global Health Discovery, WHO digital health initiatives, EU Horizon Europe Health cluster, and national ministry of health digital triage programs.

Funding Ask

\$180,000 grant to deploy SMS/USSD triage across 100 rural clinics + blood-panel screening pilot, 12 months.